2. Find the value of

(i) $\tan^{-1}\left(\sqrt{3}\right)-\sec^{-1}(-2)$ (ii) $\sin^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\cot^{-1}(2)$

(iii) $\cot^{-1}(1)+\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)-\sec^{-1}\left(-\sqrt{2}\right)$

4.10 Properties of Inverse Trigonometric Functions

In this section, we investigate some properties of inverse trigonometric functions. The properties to be discussed are valid within the principal value branches of the corresponding inverse trigonometric functions and where they are defined.

Property-I

(i) $\sin^{-1}(\sin\theta)=\theta$, if $\theta\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. (ii) $\cos^{-1}(\cos\theta)=\theta$, if $\theta\in[0,\pi]$.

(iii) $\tan^{-1}(\tan\theta)=\theta$, if $\theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$. (iv) $\text{cosec}^{1}(\text{cosec}\,\theta)=\theta$, if $\theta\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\setminus\{0\}$

(v) $\sec^{-1}(\sec\theta)=\theta$, if $\theta\in[0,\pi]\setminus\left\{\frac{\pi}{2}\right\}$. (vi) $\cot^{-1}(\cot\theta)=\theta$, if $\theta\in(0,\pi)$.

Proof

All the above results follow from the definition of the respective inverse functions.

For instance, (i) let $\sin\theta=x$; $\theta\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

Now, $\sin\theta=x$ gives $\theta=\sin^{-1}x$, by definition of inverse sine function.

Thus, $\sin^{-1}\left(\sin\theta\right)=\theta$.

Property-II

(i) $\sin\left(\sin^{-1}x\right)=x$, if $x\in[-1,1]$. (ii) $\cos\left(\cos^{-1}x\right)=x$, if $x\in[-1,1]$

(iii) $\tan\left(\tan^{-1}x\right)=x$, if $x\in\mathbb{R}$ (iv) $\text{cosec}\left(\text{cosec}^{-1}x\right)=x$, if $x\in\mathbb{R}\setminus(-1,1)$

(v) $\sec\left(\sec^{-1}x\right)=x$, if $x\in\mathbb{R}\setminus(-1,1)$ (vi) $\cot\left(\cot^{-1}x\right)=x$, if $x\in\mathbb{R}$

Proof

(i) For $x\in[-1,1]$, $\sin^{-1}x$ is well defined.

Let $\sin^{-1}x=\theta$. Then, by definition $\theta\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $\sin\theta=x$

Thus, $\sin\theta=x$ implies $\sin\left(\sin^{-1}x\right)=x$.

Similarly, other results are proved.

Note

(i) For any trigonometric function $y=f(x)$, we have $f\left(f^{-1}(x)\right)=x$ for all x in the range of f. This follows from the definition of $f^{-1}(x)$. When we have, $f\left(g^{-1}(x)\right)$, where $g^{-1}(x)=\sin^{-1}x$ or $\cos^{-1}x$, it will usually be necessary to draw a triangle defined by the inverse trigonometric function to solve the problem. For instance, to find $\cot\left(\sin^{-1}x\right)$, we have to draw a triangle using $\sin^{-1}x$. However, we have to be a little more careful with expression of the form $f^{-1}\left(f(x)\right)$.

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